Therefore, $\int \frac{1}{\sqrt{x^2+12x+11}}dx = \log\left|x+6+\sqrt{x^2+12x+11}\right|+c$

(iii) $$\int \frac{1}{\sqrt{12+4x-x^2}}dx = \int \frac{1}{\sqrt{12-(x^2-4x)}}dx$$

$$= \int \frac{1}{\sqrt{12-\{(x-2)^2-4\}}}dx$$

$$= \int \frac{1}{\sqrt{4^2-(x-2)^2}}dx$$

$$= \sin^{-1}\left(\frac{(x-2)}{4}\right)+c$$

EXERCISE 11.10

Find the integrals of the following :

(1) (i) $\frac{1}{4-x^2}$ (ii) $\frac{1}{25-4x^2}$ (iii) $\frac{1}{9x^2-4}$

(2) (i) $\frac{1}{6x-7-x^2}$ (ii) $\frac{1}{(x+1)^2-25}$ (iii) $\frac{1}{\sqrt{x^2+4x+2}}$

(3) (i) $\frac{1}{\sqrt{(2+x)^2-1}}$ (ii) $\frac{1}{\sqrt{x^2-4x+5}}$ (iii) $\frac{1}{\sqrt{9+8x-x^2}}$

Type III

Integrals of the form $\int \frac{px+q}{ax^2+bx+c}dx$ and $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$

To evaluate the above integrals, first we write

$$px+q = A\frac{d}{dx}(ax^2+bx+c)+B$$

$$px+q = A(2ax+b)+B$$

Calculate the values of A and B, by equating the coefficients of like powers of x on both sides

(i) The given first integral can be written as

$$\int \frac{px+q}{ax^2+bx+c}dx = \int \frac{A(2ax+b)+B}{ax^2+bx+c}dx$$

$$= A\int \frac{2ax+b}{ax^2+bx+c}dx + B\int \frac{1}{ax^2+bx+c}dx$$

(The first integral is of the form $\int \frac{f'(x)}{f(x)}dx$)

$$= A\log|ax^2+bx+c| + B\int \frac{1}{ax^2+bx+c}dx$$

The second term on the right hand side can be evaluated using the previous types.

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